

Numerical Analysis homework 1

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I. Length of interval

I-a

The width of the interval at the n th step is $\frac{1}{2^{n-1}}$

I-b

According to I-a, at the n th step, the distance d between the root r and the midpoint of the interval is less than half of the interval, i.e.

$$d < \frac{1}{2^{n-2}}$$

$$0 \leq \sup_n d = \sup_n \frac{1}{2^{n-2}} = 0$$

so

$$\sup_n d = 0$$

II. Accuracy guaranteed by n

Since the relative error is defined by $\frac{|c_n - r|}{r}$, where c_n is the midpoint of the interval. It suffices to prove that

$$\frac{|c_n - r|}{r} < \epsilon \quad \text{when} \quad n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1$$

Taking logarithms on the both side of the last equation, we obtain

$$\epsilon \geq \frac{b_0 - a_0}{2^{n+1}a_0}$$

From Problem I, we know that $|c_n - \alpha| < \frac{1}{2} \text{Length of the interval} = \frac{b_0 - a_0}{2^{n+1}}$

Now, since $a_0 < r$, we have

$$\frac{|c_n - \alpha|}{r} < \frac{b_0 - a_0}{2^{n+1}a_0} \leq \epsilon$$

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III. Newton's method by hand calculator

We apply Newton's method with initial guess $x_0 = -1$. The iteration formula is:

$$x_{n+1} = x_n - \frac{p(x_n)}{p'(x_n)}$$

where $p(x) = 4x^3 - 2x^2 + 3$ and $p'(x) = 12x^2 - 4x$.

We perform four iterations. The results are summarized in the table below. Values are rounded to six decimal places for clarity.

Iteration n	x_n	$p(x_n)$	$p'(x_n)$	x_{n+1}
0	-1.000000	-3.000000	16.000000	-0.812500
1	-0.812500	-0.465820	11.171875	-0.770805
2	-0.770805	-0.020145	10.212906	-0.768832
3	-0.768832	-0.000056	10.168552	-0.768827

Table 1: Newton's method iterations for $p(x) = 4x^3 - 2x^2 + 3$

IV. A variation of Newton's method

By Taylor's theorem,

$$f(\alpha) = f(x_n) + (\alpha - x_n)f'(\xi)$$

where ξ is between α and x_n . Then $f(\alpha) = 0$ gives

$$f(x_n) = (x_n - \alpha)f'(\xi)$$

Now, since $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$, we replace $f(x_n)$ by $(x_n - \alpha)f'(\xi)$ and yield

$$x_{n+1} - \alpha = x_n - \alpha - (x_n - \alpha)\frac{f'(\xi)}{f'(x_0)}$$

Let $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} = \lim_{n \rightarrow \infty} \frac{x_{n+1} - \alpha}{x_n - \alpha} = 1 - \frac{f'(\alpha)}{f'(x_0)}$$

i.e.

$$s = 1, C = 1 - \frac{f'(\alpha)}{f'(x_0)}$$

V. Convergence of iteration

To analyze convergence, I discuss it in three categories.

Case 1: $0 < x_0 < \frac{\pi}{2}$

Since $\tan^{-1}(x) < x$ for all $x > 0$, the sequence $\{x_n\}$ is strictly decreasing. Moreover, it is bounded below by 0 because $\tan^{-1}(x) > 0$ for $x > 0$. By the Monotone Convergence Theorem, the sequence converges to some limit $L \geq 0$. Taking the limit in the recurrence $x_{n+1} = \tan^{-1}(x_n)$, and using continuity of f , we get $L = \tan^{-1}(L)$. The only solution when $0 < L < \frac{\pi}{2}$ is $L = 0$. Hence, the sequence converges to 0.

Case 2: $x_0 < 0$

For $x < 0$, we have $\tan^{-1}(x) > x$, so the sequence is strictly increasing. It is bounded above by 0 because $\tan^{-1}(x) < 0$ for $x < 0$. Again, by monotonicity and boundedness, the sequence converges to a limit $L \leq 0$, which must satisfy $L = \tan^{-1}(L)$. The only solution when $-\frac{\pi}{2} < L < 0$ is $L = 0$. Thus, convergence to 0 occurs.

Case 3: $x_0 = 0$

Then the sequence is constant at 0.

Therefore, for any $x_0 \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the iteration $x_{n+1} = \tan^{-1}(x_n)$ converges to 0.

VI. Continued fraction

Let $x = \lim_{n \rightarrow \infty} x_n$, where $x_1 = \frac{1}{p}, x_2 = \frac{1}{p + \frac{1}{p}}, \dots$, obviously, $x_1 > x_3 > x_5 > \dots > \frac{1}{p+1}, x_2 < x_4 < \dots < 1$

Now we have the recursion formula that

$$x_n = \frac{1}{x_{n+1}} - p$$

Thus

$$x_n = \frac{1}{\frac{1}{x_{n+2}} - p} - p$$

For $a_n = x_{2n+1}, n = 0, 1, 2, \dots$

By the Monotone Convergence Theorem, the sequence a_n converges to some limit $L \geq 0$

Let $L = \frac{1}{L} - p$, the only solution satisfies $L > 0$ is $L = \frac{\sqrt{p^2 + 4} - p}{2}$

For $b_n = x_{2n}, n = 1, 2, 3, \dots$

Similarly, the only solution is $L = \frac{\sqrt{p^2 + 4} - p}{2}$.

Therefore, $x = \lim_{n \rightarrow \infty} x_n = \frac{\sqrt{p^2 + 4} - p}{2}$

VII. Another situation of Problem II

The definition of relative error is $\frac{|c_n - r|}{|r|} (r \neq 0)$, where c_n is the midpoint of the n th interval, r is the root.

By method in Problem II, we only need to find n that satisfies:

$$\frac{|c_n - r|}{|r|} < \frac{b_0 - a_0}{2^{n+1}r} < \epsilon$$

Take logarithms on the both side

$$n > \frac{\log(b_0 - a_0) - \log \epsilon - \log |r|}{\log 2} - 1$$

Since $|r|$ is unknown, we cannot find a lower bound for $|r|$, and therefore cannot further simplify this expression.

VIII. Function f with a zero of multiplicity k

VIII-a

If α is a zero of multiplicity k of the function f , $f^{(i)}(\alpha) = 0, i = 0, 1, \dots, k-1$

By Taylor's theorem,

$$f(x) = f(\alpha) + \dots + \frac{f^{(k)}(\alpha)}{k!}(x - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x - \alpha)^{k+1} = \frac{f^{(k)}(\alpha)}{k!}(x - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x - \alpha)^{k+1}$$

Similarly,

$$f'(x) = \frac{f^{(k)}(\alpha)}{(k-1)!}(x - \alpha)^{k-1} + \frac{f^{(k+1)}(\xi)}{k!}(x - \alpha)^k$$

Because $x_{n+1} - \alpha = x_n - \alpha - \frac{f(x_n)}{f'(x_n)}$

$$\frac{x_{n+1} - \alpha}{x_n - \alpha} = 1 - \frac{\frac{f^{(k)}(\alpha)}{k!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x_n - \alpha)^{k+1}}{\frac{f^{(k)}(\alpha)}{(k-1)!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi)}{k!}(x_n - \alpha)^{k+1}} = 1 - \frac{f^{(k)}(\alpha) + \frac{f^{(k+1)}(\xi)}{k+1}(x_n - \alpha)}{kf^{(k)}(\alpha) + f^{(k+1)}(\xi)(x_n - \alpha)}$$

Let $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - \alpha}{x_n - \alpha} = 1 - \frac{1}{k}$$

Hence

$$\frac{f(x_{n+1})}{f(x_n)} = \frac{\frac{f^{(k)}(\alpha)}{k!}(x_{n+1} - \alpha)^k + \frac{f^{(k+1)}(\xi_1)}{(k+1)!}(x_{n+1} - \alpha)^{k+1}}{\frac{f^{(k)}(\alpha)}{k!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi_2)}{(k+1)!}(x_n - \alpha)^{k+1}} = \frac{(k+1)f^{(k)}(\alpha)\left(\frac{x_{n+1} - \alpha}{x_n - \alpha}\right)^k + f^{(k+1)}(\xi_1)\left(\frac{x_{n+1} - \alpha}{x_n - \alpha}\right)^k(x_{n+1} - \alpha)}{(k+1)f^{(k)}(\alpha) + f^{(k+1)}(\xi_2)(x_n - \alpha)}$$

Let $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \frac{f(x_{n+1})}{f(x_n)} = \left(1 - \frac{1}{k}\right)^k$$

That is, a multiple zero can be detected by calculating $\frac{f(x_{n+1})}{f(x_n)}$ for a sufficiently large iteration index n .

VIII-b

Let $f(x) = (x - r)^k g(x)$, so $g(r), g'(r) \neq 0$.

By the modification,

$$\begin{aligned} |e_{n+1}| &= |x_{n+1} - r| \\ &= \left| x_n - r - k \frac{(x_n - r)^k g(x_n)}{k(x_n - r)^{k-1} g(x_n) + (x_n - r)^k g'(x_n)} \right| \\ &= \left| e_n - e_n \frac{g(x_n)}{g(x_n) + \frac{x_n - r}{k} g'(x_n)} \right| \\ &= \left| e_n^2 \frac{g'(x_n)}{kg(x_n) + e_n g'(x_n)} \right| \end{aligned}$$

i.e.

$$\left| \frac{e_{n+1}}{e_n^2} \right| = \frac{g'(r)}{kg(r)} > 0 (n \rightarrow \infty)$$